

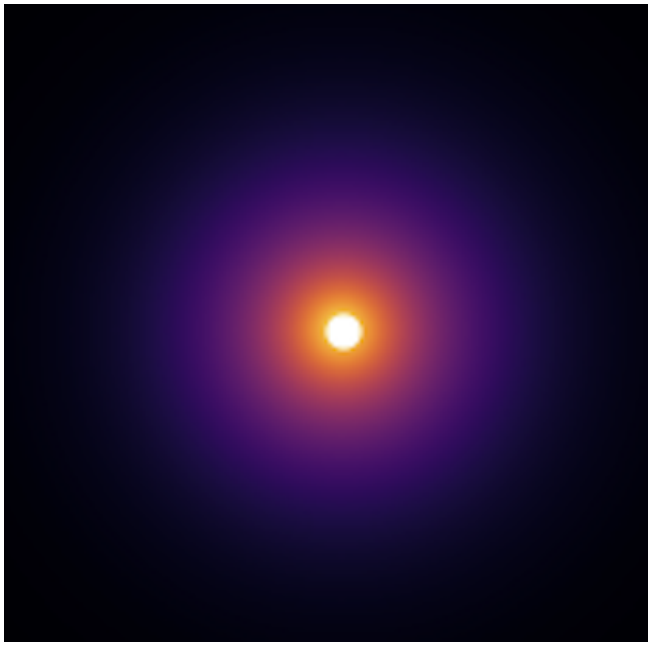
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Final Report

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Author: Ava F. Piper

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Faculty of Science
The University of Melbourne

Subject co-ordinators: A/Prof. Suzie Sheehy & Prof. Christian Reichardt

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Chapter 1

Introduction

1.1 Hydrogenic atoms

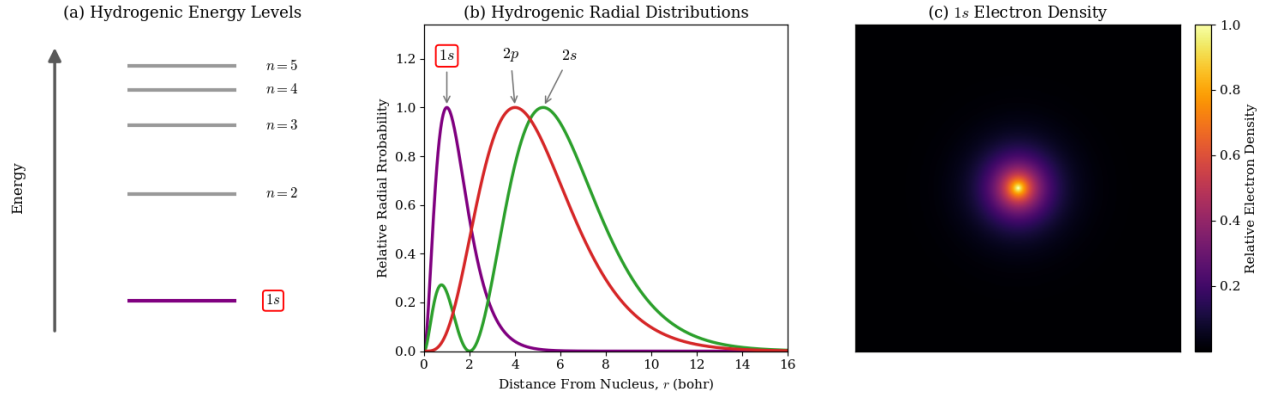


Figure 1.1: Caption

Hydrogenic atoms are atomic systems containing a single electron bound to a nucleus of charge $+Ze$, where Z is the atomic number. The simplest example is hydrogen ($Z = 1$), although the same description applies to one-electron ions such as He^{+}, Li^{2+}, and Mn^{24+}.

Unlike multi-electron atoms, hydrogenic systems contain no electron–electron interactions. Their nonrelativistic Schrödinger equation can therefore be solved analytically, providing exact wavefunctions and energy eigenvalues against which numerical methods can be tested. *(Insert Bransden and Joachain reference upon revision) ...*

In atomic units, and assuming an infinitely massive point nucleus, the Hamiltonian is

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{Z}{r} \tag{1.1}$$

where r is the electron–nucleus separation. The corresponding ground-state energy and normalised $1s$ wavefunction are

$$E_{1s} = -\frac{Z^2}{2}, \qquad \psi_{1s}(r) = \sqrt{\frac{Z^3}{\pi}} e^{-Zr} \tag{1.2}$$

The exponential dependence describes the spatial decay of the electron wavefunction away from the nucleus. As Z increases, the electron becomes more tightly bound and its characteristic spatial extent decreases.

Hydrogenic systems are consequently useful benchmarks for investigating the accuracy, convergence and numerical stability of approximate quantum-mechanical methods.

1.2 Gaussian basis sets

Although the hydrogenic Schrödinger equation has an analytical solution, most atomic and molecular systems cannot be solved exactly. Numerical electronic-structure methods therefore commonly approximate wavefunctions using finite expansions of simpler basis functions.

One widely used approach employs Gaussian-type orbitals , which were introduced into electronic-structure calculations by Boys [2].

A spherical Gaussian basis function has the form

$$\phi_i(r) = A_i e^{-\lambda_i r^2} \tag{1.3}$$

where $\lambda_i > 0$ determines the spatial width and A_i is a normalisation constant.

Small exponents produce broad functions extending farther from the nucleus, whereas large exponents produce narrow functions concentrated near the nucleus.

The hydrogenic ground-state wavefunction can be approximated by a linear combination of N Gaussian functions,

$$\psi_{\mathrm{G}}(r) = \sum_{i=1}^{N} c_i \phi_i(r) \tag{1.4}$$

where the coefficients c_i determine the contribution of each basis function.

The coefficients are obtained by minimising the energy expectation value, leading to a generalised matrix eigenvalue problem. The resulting eigenvalues provide approximate energies, while the eigenvectors determine the corresponding wavefunctions.

The principal computational advantage of Gaussian functions is that their products and many associated integrals can be evaluated analytically. This makes them particularly convenient for constructing the Hamiltonian and overlap matrices required in electronic-structure calculations [2].

Gaussian basis sets have consequently become a standard component of quantum-chemical calculations, including more elaborate contracted and correlation- consistent basis sets [3, 8].

1.3 Accuracy and limitations of Gaussian expansions

Despite their computational advantages, Gaussian functions do not have the same radial dependence as exact hydrogenic orbitals.

The exact $1s$ wavefunction decays as e^{-Zr}, whereas an individual Gaussian decays as $e^{-\lambda r^2}$. A finite combination of Gaussians can approximate the exponential function closely over a range of distances, but cannot reproduce its behaviour exactly everywhere.

An important difference occurs at the nucleus. The exact hydrogenic wavefunction satisfies the electron–nucleus cusp condition,

$$\left.\frac{\mathrm{d}\psi_{1s}}{\mathrm{d}r}\right|_{r=0} = -Z\psi_{1s}(0) \tag{1.5}$$

whereas the radial derivative of any finite combination of centred spherical Gaussians vanishes at $r = 0$.

This limitation is connected to the Coulomb singularity and the behaviour of electronic wavefunctions near nuclei [5]. The theoretical accuracy of Gaussian expansions has also been investigated by Kutzelnigg [6].

Increasing the number of basis functions generally allows a more flexible approximation, although the rate of improvement depends on the chosen Gaussian exponents.

In particular, a basis containing functions with very similar widths may become nearly linearly dependent, leading to an ill-conditioned overlap matrix and potential numerical instability [7].

The selection of Gaussian exponents is therefore as important as the number of basis functions.

1.4 Even-tempered Gaussian basis sets

Here the term *even-tempered* refers to the constant ratio between successive Gaussian exponents, i.e. $\lambda_{i+1}/\lambda_i = \beta$, meaning that the exponents are equally spaced on a logarithmic scale. A systematic way of selecting Gaussian exponents is to use a geometric progression,

$$\lambda_i = \alpha\beta^{i-1}, \qquad i = 1, \ldots, N \tag{1.6}$$

where $\alpha > 0$ sets the smallest exponent, $\beta > 1$ controls the spacing between successive exponents, and N is the number of functions.

These are commonly referred to as even-tempered Gaussian basis sets. Their systematic construction makes it possible to investigate the influence of basis size and exponent distribution independently.

Even-tempered bases and the optimisation of Gaussian exponents have been studied extensively by Feller and Ruedenberg [4], Bardo and Ruedenberg [1], and Wilson [9].

For the present investigation, the parameters N, α, and β provide a convenient means of controlling the flexibility and numerical behaviour of the basis.

1.5 Motivation and scope

The hydrogenic ground state provides a useful test case because both its energy and wavefunction are known analytically. This allows numerical accuracy to be assessed without uncertainty arising from the reference solution.

The present investigation considers not only the convergence of the ground-state energy, but also the accuracy of the reconstructed wavefunction and radial probability distribution.

This distinction is important because a small energy error does not necessarily imply an equally small error in every observable or in the local behaviour of the wavefunction [6].

A further question concerns the transferability of a Gaussian basis between hydrogenic systems. Increasing nuclear charge contracts the exact wavefunction, suggesting that a basis optimised for hydrogen may be unsuitable for more strongly bound one-electron ions.

The dependence of hydrogenic length scales on nuclear charge provides a natural way to investigate whether Gaussian exponents can be rescaled to preserve accuracy.

1.6 Aims and objectives

The aim of this report is to investigate the accuracy, convergence and transferability of finite Gaussian basis approximations to hydrogenic ground states.

The principal objectives are:

1. Construct a Gaussian basis approximation to the hydrogen $1s$ wavefunction and compare it with the exact solution.
2. Investigate how the ground-state energy error depends on the number of Gaussian basis functions.
3. Examine the influence of the exponent parameters α and β.
4. Assess numerical stability through the condition number of the overlap matrix.
5. Compare energy accuracy with wavefunction and radial-observable accuracy.
6. Investigate the transferability of a hydrogen basis to higher Z hydrogenic ions, up to and including Mn^{24+}.
7. Examine whether nuclear-charge scaling of the Gaussian exponents preserves the accuracy of the approximation.

1.7 Report structure

Chapter 2: Introduces the theoretical framework, including Gaussian normalisation, the variational method, matrix elements and accuracy measures.

Chapter 3: Describes the numerical implementation and the parameter investigations.

Chapter 4: Presents the results for hydrogen, basis-set convergence, exponent selection, numerical conditioning and nuclear-charge scaling.

Chapter 5: Summarises the main findings, limitations and possible extensions.

Chapter 2

Theory

2.1 The hydrogenic Schrödinger equation

$$\hat{H}\psi = E\psi \tag{2.1}$$

2.2 Gaussian basis representation

2.2.1 Normalisation of Gaussian functions

$$\phi_i(r) = \left(\frac{2\lambda_i}{\pi}\right)^{3/4} e^{-\lambda_i r^2} \tag{2.2}$$

2.2.2 Linear combination of Gaussians

$$\psi_{\mathrm{G}}(r) = \sum_{i=1}^{N} c_i \phi_i(r) \tag{2.3}$$

2.2.3 Even-tempered exponents

$$\lambda_i = \alpha\beta^{i-1}, \qquad i = 1, \ldots, N \tag{2.4}$$

2.3 Variational method

2.3.1 Energy expectation value

$$E[\psi] = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} \tag{2.5}$$

$$E_{\mathrm{G}} \geq E_{\mathrm{exact}} \tag{2.6}$$

2.3.2 Generalised eigenvalue problem

$$H\mathbf{c} = ES\mathbf{c} \tag{2.7}$$

2.4 Hamiltonian and overlap matrices

2.4.1 Overlap integrals

$$S_{ij} = \langle\phi_i|\phi_j\rangle \tag{2.8}$$

Writing

$$p = \lambda_i + \lambda_j, \qquad A_i = \left(\frac{2\lambda_i}{\pi}\right)^{3/4} \tag{2.9}$$

gives

$$S_{ij} = A_i A_j \left(\frac{\pi}{p}\right)^{3/2} \tag{2.10}$$

2.4.2 Kinetic-energy integrals

$$T_{ij} = \left\langle \phi_i \left| -\frac{1}{2}\nabla^2 \right| \phi_j \right\rangle \tag{2.11}$$

$$T_{ij} = S_{ij}\frac{3\lambda_i\lambda_j}{p} \tag{2.12}$$

2.4.3 Potential-energy integrals

$$V_{ij} = \left\langle \phi_i \left| -\frac{Z}{r} \right| \phi_j \right\rangle \tag{2.13}$$

$$V_{ij} = -ZA_iA_j\frac{2\pi}{p} \tag{2.14}$$

2.4.4 Hamiltonian matrix

$$H_{ij} = T_{ij} + V_{ij} \tag{2.15}$$

2.5 Measures of accuracy

2.5.1 Ground-state energy error

$$\epsilon_E = \frac{|E_\mathrm{G} - E_\mathrm{exact}|}{|E_\mathrm{exact}|} \tag{2.16}$$

2.5.2 Wavefunction and radial probability

$$P(r) = 4\pi r^2 |\psi(r)|^2 \tag{2.17}$$

$$\int_0^\infty P(r)\,\mathrm{d}r = 1 \tag{2.18}$$

2.5.3 Mean electron radius

$$\langle r\rangle = \int_0^\infty rP(r)\,\mathrm{d}r \tag{2.19}$$

For the exact ground state,

$$\langle r\rangle_\mathrm{exact} = \frac{3}{2Z} \tag{2.20}$$

2.5.4 Wavefunction stuff

$$F = |\langle\psi_\mathrm{G}|\psi_\mathrm{exact}\rangle|^2 \tag{2.21}$$

2.6 Numerical conditioning

$$\kappa_2(S) = \frac{s_\mathrm{max}}{s_\mathrm{min}} \tag{2.22}$$

2.7 Electron–nucleus cusp

$$\psi'_{1s}(0) = -Z\psi_{1s}(0) \tag{2.23}$$

$$\psi'_\mathrm{G}(0) = 0 \tag{2.24}$$

2.8 Hydrogenic nuclear-charge scaling

$$\rho = Zr \tag{2.25}$$

$$\psi_Z(r) = Z^{3/2}\psi_1(Zr) \tag{2.26}$$

$$\lambda_i(Z) = Z^2\lambda_i(1) \tag{2.27}$$

$$E_{\mathrm{G}}(Z) = Z^2 E_{\mathrm{G}}(1) \tag{2.28}$$

Chapter 3

Computational Method

3.1 Numerical implementation

3.2 Construction of the Gaussian basis

3.3 Matrix construction

3.4 Solving the eigenvalue problem

3.5 Algorithm flowchart

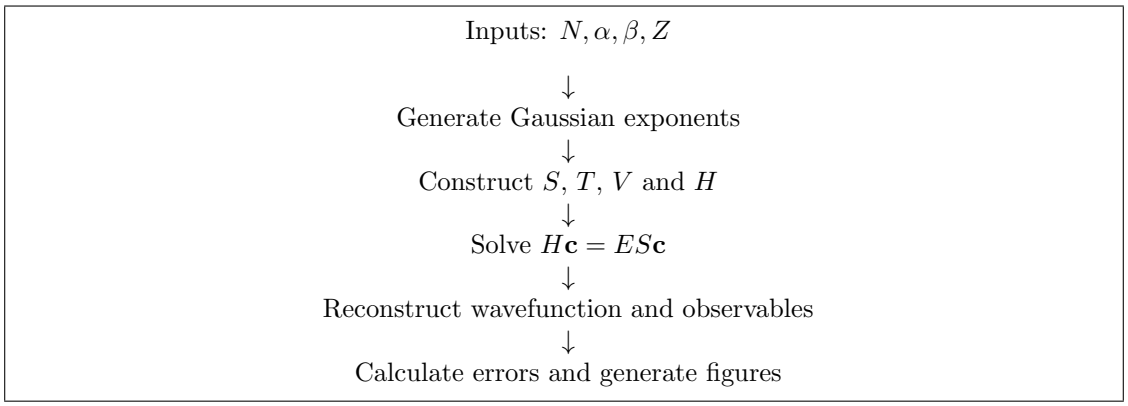


Figure 3.1: Computational procedure for the Gaussian basis calculations.

3.6 Hydrogen benchmark

3.7 Basis-size convergence

3.8 Exponent parameter investigation

3.9 Nuclear-charge investigation

3.10 Hydrogenic manganese

3.11 Dimensionless radial scaling

3.12 Numerical verification

Chapter 4

Results and Discussion

4.1 Hydrogen ground-state approximation

Table 4.1: Hydrogen ground-state benchmark using twelve Gaussian basis functions.

Quantity	Exact	Gaussian
Energy (Ha)	-0.500000	-0.499589
Relative energy error	0	8.2170×10^{-4}
Mean radius (bohr)	1.500000	1.501244
Condition number	–	8.3295×10^{4}

4.1.1 Wavefunction comparison

4.1.2 Radial probability distribution

4.2 Convergence with basis size

4.3 Dependence on Gaussian exponents

4.4 Energy and wavefunction accuracy

4.5 Transferability across nuclear charge

4.6 Hydrogenic manganese

Table 4.2: Ground-state energies for hydrogenic manganese ($Z = 25$).

Method	Energy (Ha)	Relative error
Exact	-312.500000	0
Fixed hydrogen basis	-140.956593	0.54894
Z^2-scaled basis	-312.243220	8.2170×10^{-4}

4.7 Dimensionless scaling

4.8 Practical basis-set selection

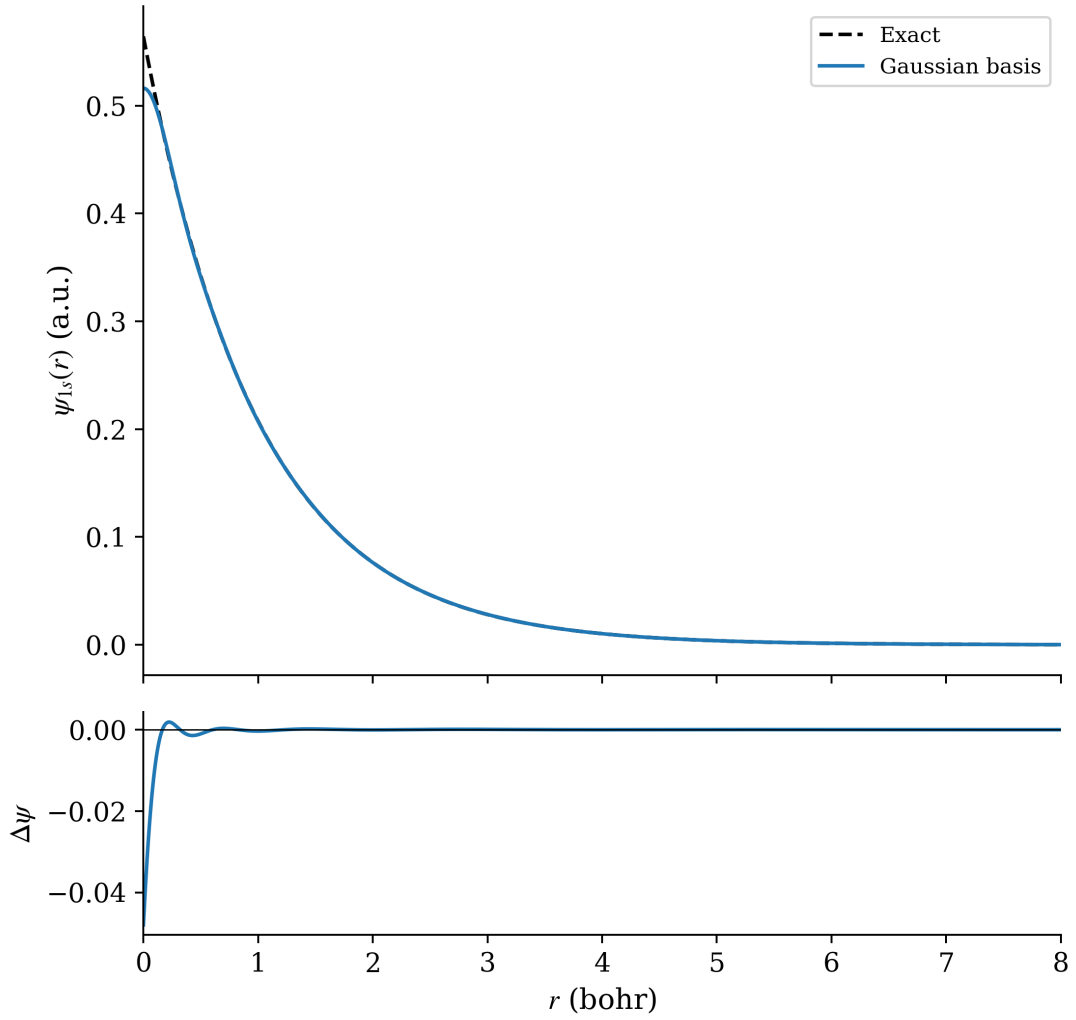


Figure 4.1: Comparison of the exact hydrogen ground-state wavefunction and its Gaussian basis approximation, with the wavefunction residual shown below.

Table 4.3: Basis sizes required to reach selected relative energy-error targets.

Target	N	α	β	$\kappa_2(S)$
10^{-3}	–	–	–	–
10^{-4}	–	–	–	–
10^{-5}	–	–	–	–
10^{-6}	–	–	–	–

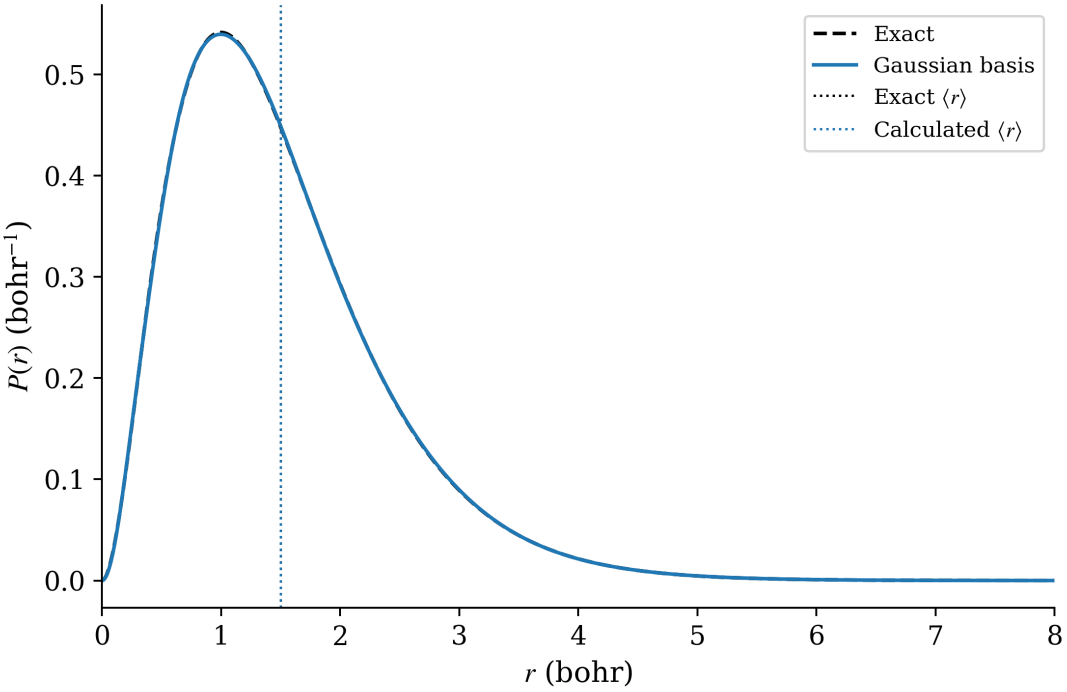


Figure 4.2: Exact and Gaussian radial probability distributions for hydrogen, including the corresponding mean electron radii.

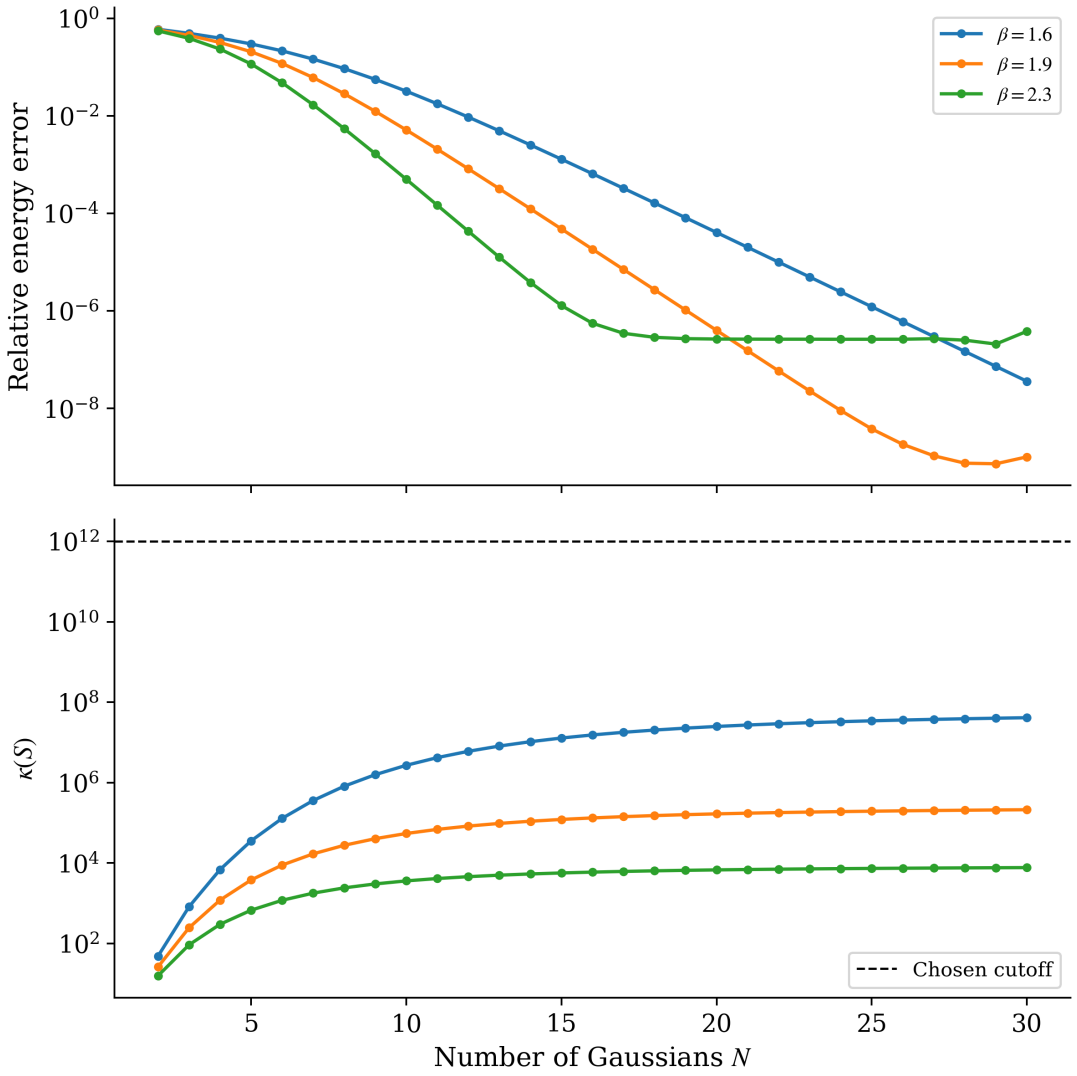


Figure 4.3: Relative ground-state energy error and overlap-matrix condition number versus the number of Gaussian basis functions, for three exponent spacings.

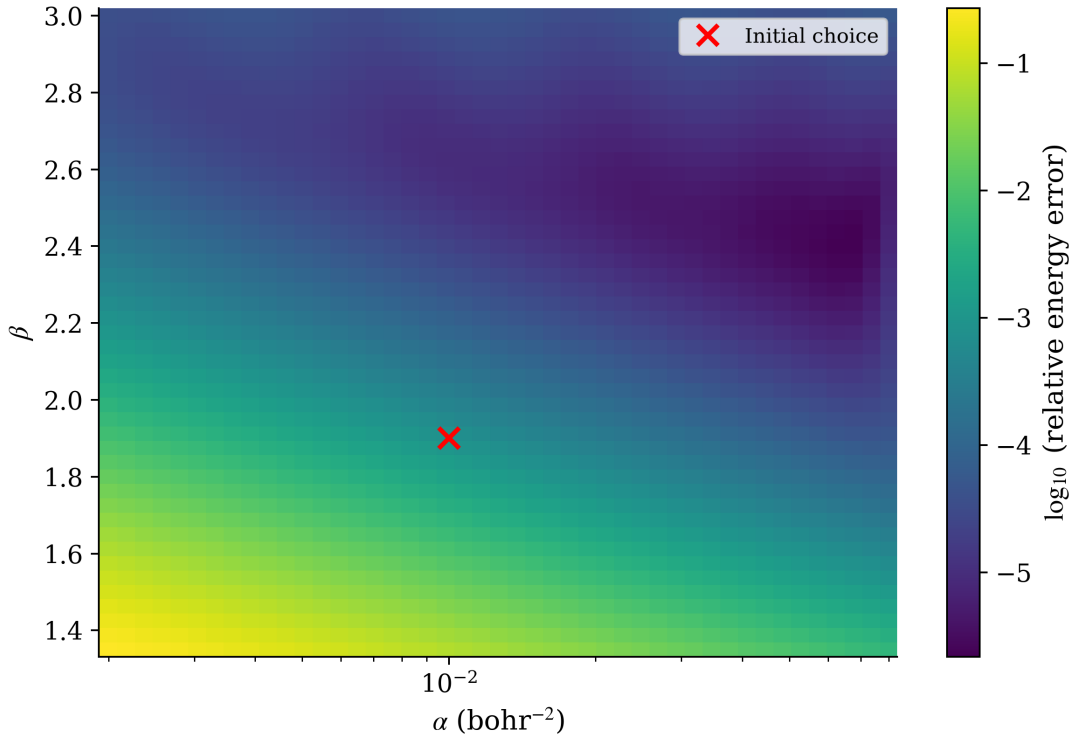


Figure 4.4: Logarithm of the relative ground-state energy error as a function of the Gaussian exponent parameters α and β, with the initial parameter choice indicated.

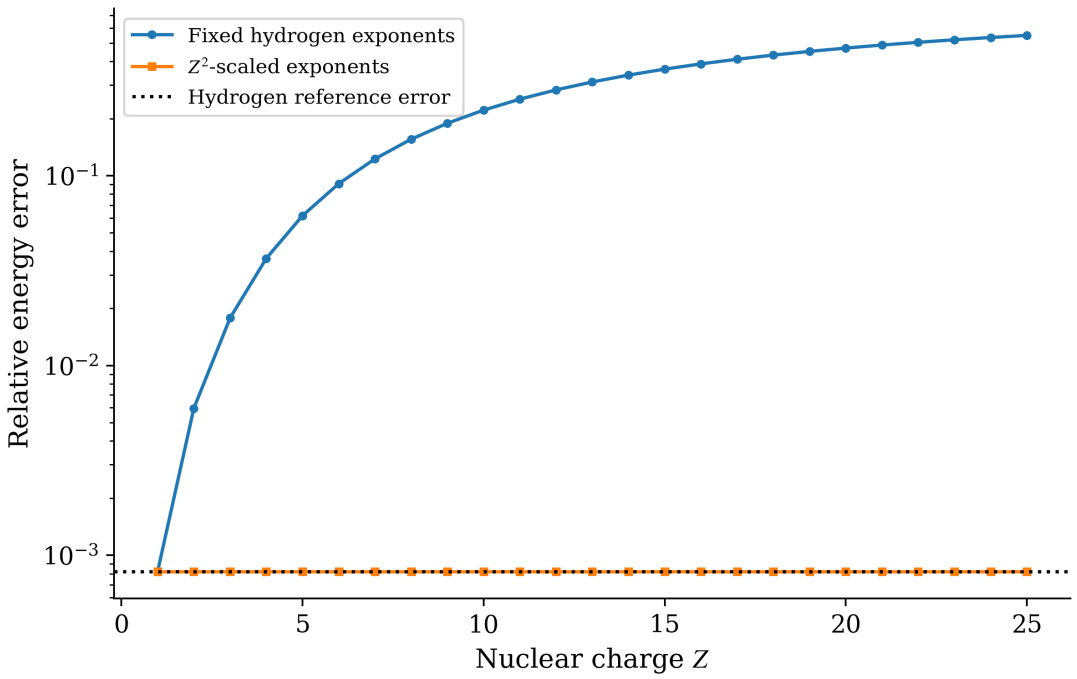


Figure 4.5: Relative ground-state energy error for hydrogenic ions with $1 \leq Z \leq 25$, comparing fixed hydrogen exponents with Z^2-scaled exponents.

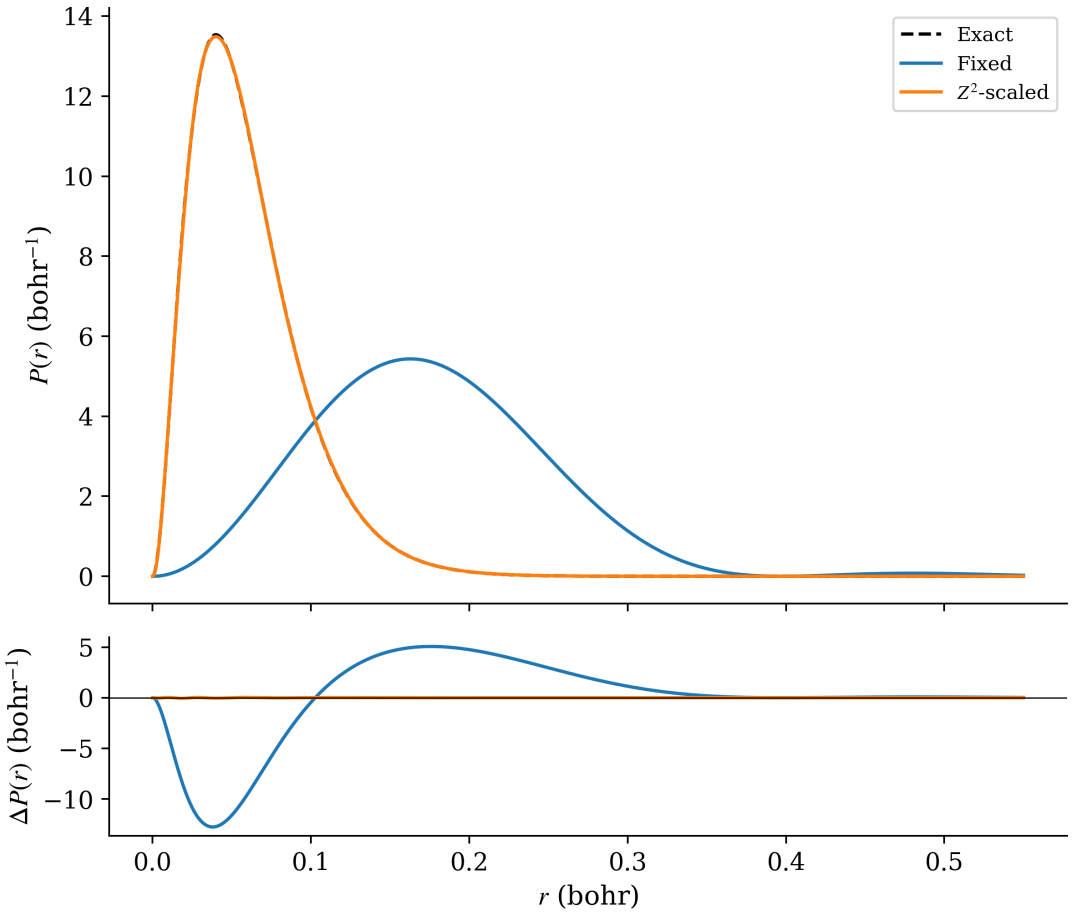


Figure 4.6: Exact and approximate radial probability distributions for hydrogenic manganese, comparing fixed and Z^2-scaled Gaussian bases.

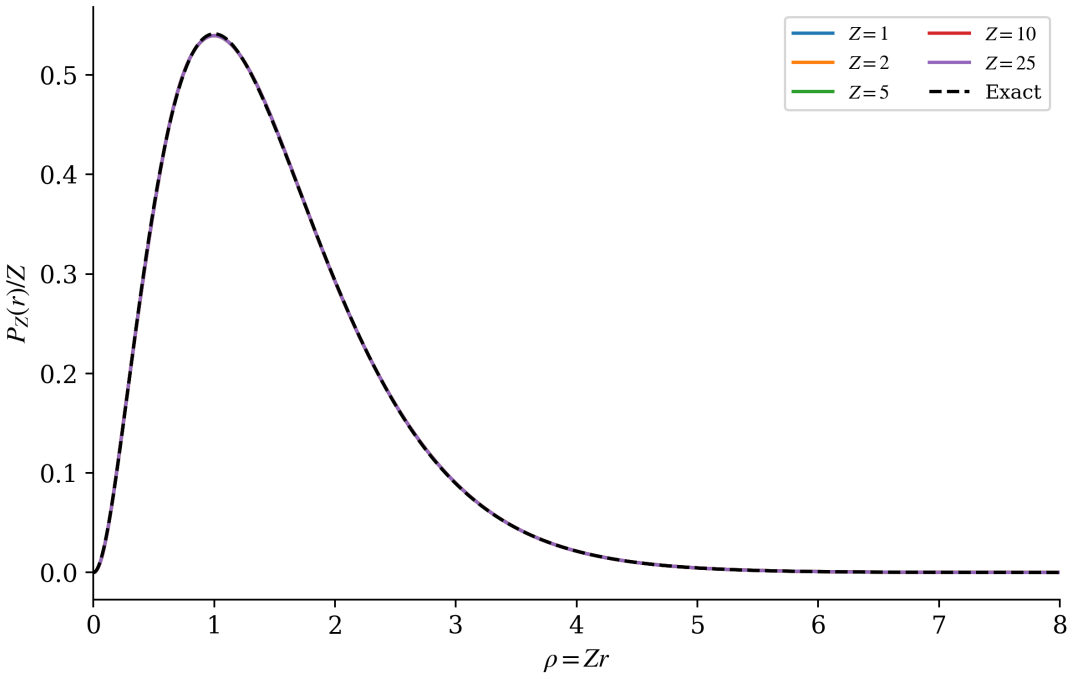


Figure 4.7: Scaled radial probability distributions for selected nuclear charges, plotted against the dimensionless coordinate $\rho = Zr$.

Chapter 5

Conclusion and Future Work

5.1 Summary of findings

5.2 Limitations

5.3 Future work

5.4 Final remarks

Appendix A

Appendix

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